

Lecture 11: Overfitting

Causes, Mechanisms, and Diagnostic Indicators

Caltech CS156: Learning from Data

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1 Definition of Overfitting

Definition: Overfitting

Overfitting occurs when a model fits the in-sample training data too closely, capturing noise or spurious patterns rather than the underlying target function, thereby deteriorating out-of-sample performance E_{out} .

Overfitting is a relative phenomenon: Model $\mathcal{H}_B$ overfits relative to $\mathcal{H}_A$ if $E_{\text{in}}(\mathcal{H}_B) < E_{\text{in}}(\mathcal{H}_A)$ but $E_{\text{out}}(\mathcal{H}_B) > E_{\text{out}}(\mathcal{H}_A)$.

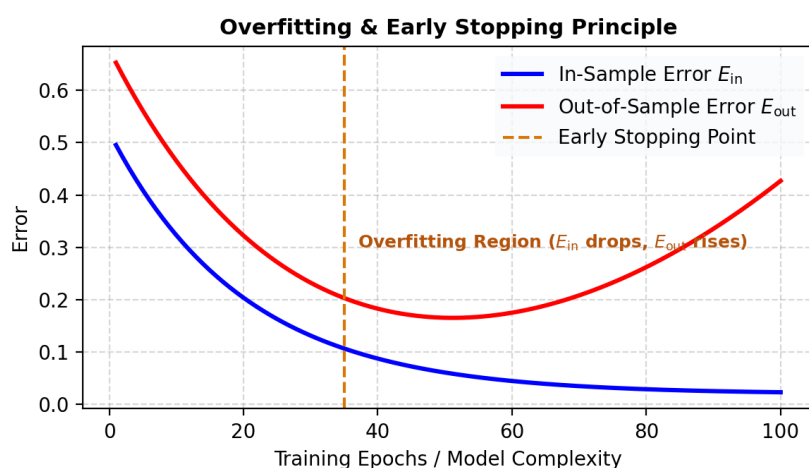


Figure 1: Overfitting trajectory as model complexity or training epochs increase.

2 The Two Sources of Noise

Overfitting is primarily caused by fitting **Noise**. Lecture 11 establishes that noise consists of two distinct components:

1. **Stochastic Noise** (σ^2): Random measurement errors, corruption, or non-deterministic fluctuations inherent in the data generation process $y = f(\mathbf{x}) + \epsilon$.
2. **Deterministic Noise**: The portion of the true target function $f(\mathbf{x})$ that the chosen hypothesis set $\mathcal{H}$ cannot capture due to restricted complexity.

Decomposition of Target Error

For any hypothesis set $\mathcal{H}$ and best approximating hypothesis $h^* \in \mathcal{H}$:

$$y - h^*(\mathbf{x}) = \underbrace{(y - f(\mathbf{x}))}_{\text{Stochastic Noise } \epsilon} + \underbrace{(f(\mathbf{x}) - h^*(\mathbf{x}))}_{\text{Deterministic Noise}} \quad (1)$$

3 Experimental Overfitting Heatmaps

Systematic experiments varying sample size N , stochastic noise σ^2 , and target complexity Q_f reveal:

- Overfitting increases as sample size N decreases.
- Overfitting increases as stochastic noise σ^2 increases.
- Overfitting increases as target complexity Q_f (deterministic noise) increases.

4 Remedies Against Overfitting

To mitigate overfitting in machine learning workflows, two primary methodologies are used:

1. **Regularization** ("Putting the brakes"): Constraining model capacity during optimization to prevent parameters from fitting noise.
2. **Validation** ("Checking the bottom line"): Estimating E_{out} on an independent holdout set to guide model selection and trigger **Early Stopping**.